

Exercises II : Topology III

August 28, 2025

1. Let $Y \subset X$. Prove that the image of $\pi_2(X)$ in $\pi_2(X, Y)$ lies in the centre of the group.
2. Let $Y \subset X$ be a subcomplex of a CW complex X . Suppose that there is a retraction $r : X \rightarrow Y$. Prove that there is a map from $\Sigma X \rightarrow \Sigma Y \vee \Sigma(X/Y)$ which is a homology isomorphism.
3. Construct a map $\Sigma(X \times Y) \rightarrow \Sigma X \vee \Sigma Y \vee \Sigma(X \wedge Y)$ which induces an isomorphism in homology. [Recall that $[\Sigma Z, W]_*$ has a group structure.]
4. Prove that $\mathbb{R}P^n/\mathbb{R}P^{n-2}$ is homotopy equivalent to $S^n \vee S^{n-1}$ if n is odd, and $\Sigma^{n-2}(\mathbb{R}P^2)$ if n is even.
5. Prove that $L_{2n+1}(C_m)/L_{2n-1}(C_m)$ is homotopy equivalent to $S^{2n} \vee S^{2n+1}$. ($L_{2n+1}(C_m)$ is the Lens space S^{2n+1}/C_m)
6. Let X be n -connected, Y be m -connected. Then, prove that the inclusion $X \vee Y \rightarrow X \times Y$ is a $(m+n+1)$ -equivalence. Conclude that for $m, n \geq 2$,

$$\pi_r(S^n \vee S^m) \cong \pi_r(S^n) \oplus \pi_r(S^m), \text{ if } r \leq m+n-2.$$

7. Suppose that X, Y, Z are spaces, and define the space $F_2(X, Y, Z) \subset X \times Y \times Z$ as

$$F_2(X, Y, Z) = \{(x, y, z) \in X \times Y \times Z \mid \text{at least one of } x, y, z \text{ is } *\}.$$

Prove that the inclusion $F_2(X, Y, Z) \rightarrow X \times Y \times Z$ is surjective on homotopy groups.

8. **a)** Let $M(\mathbb{Z}/k, n)$ be the Moore space defined as the mapping cone $S^n \xrightarrow{k} S^n$, where k stands for the degree k map. Prove that $\pi_n M(\mathbb{Z}/k, n) \cong \mathbb{Z}/k$.
b) Let $\alpha \in \pi_n(X)$ for $n \geq 2$ such that $k\alpha = 0$. Prove that there is a map $\phi : M(\mathbb{Z}/k, n) \rightarrow X$ such that $\phi_*([1]) = \alpha$ on π_n .
9. Recall that $\simeq$ is the relation on the collection of topological spaces generated by weak equivalences. Prove that $X \simeq Y$ if and only if there is a CW complex Z with weak equivalences $Z \rightarrow X$ and $Z \rightarrow Y$.